



# Generative AI Academy

# Function calling and Agents

## Query BFSI customer information

This web API presents an interface to ask questions on customer information stored in a database.

user\_input

Do customers who default maintain a low balance?

Clear

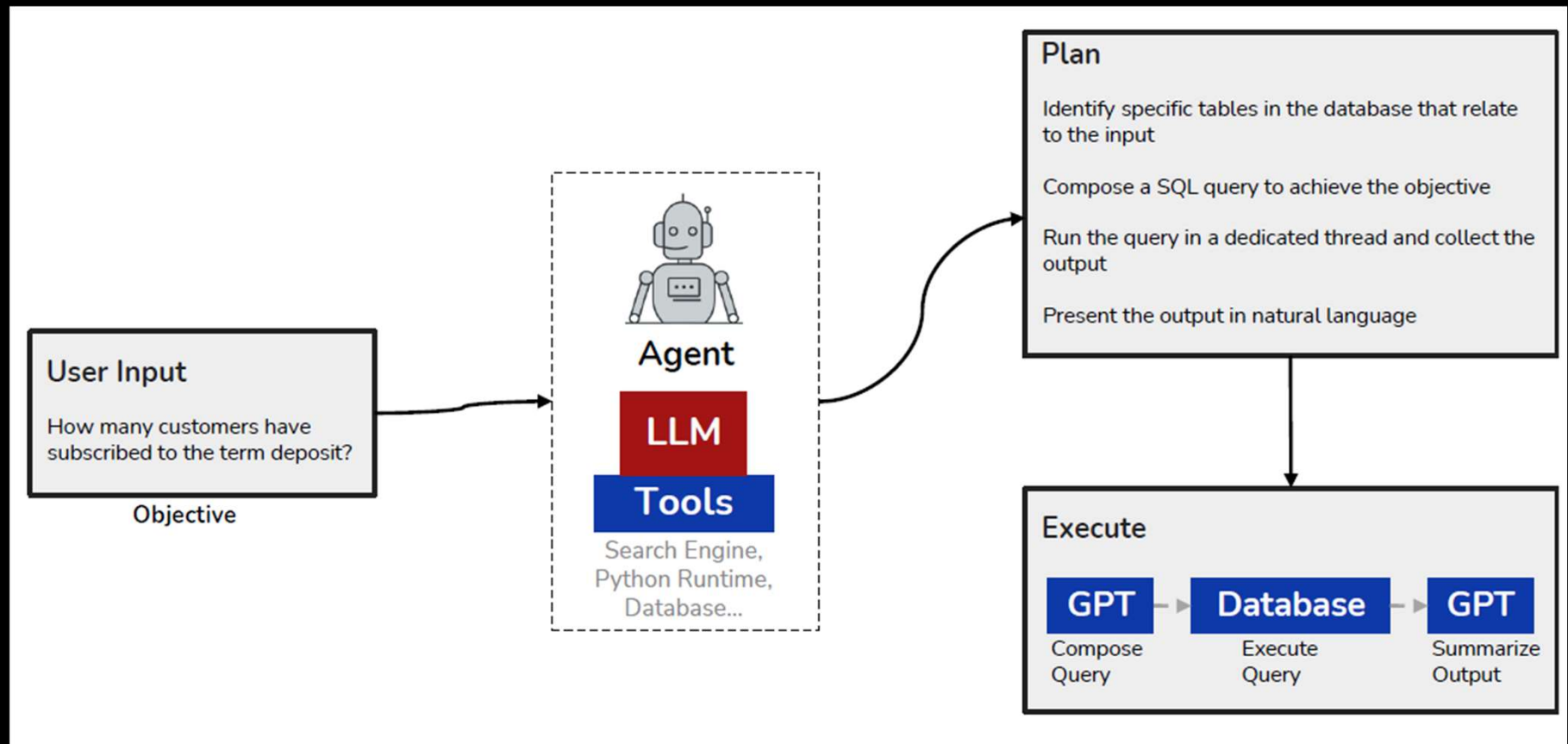
Submit

output

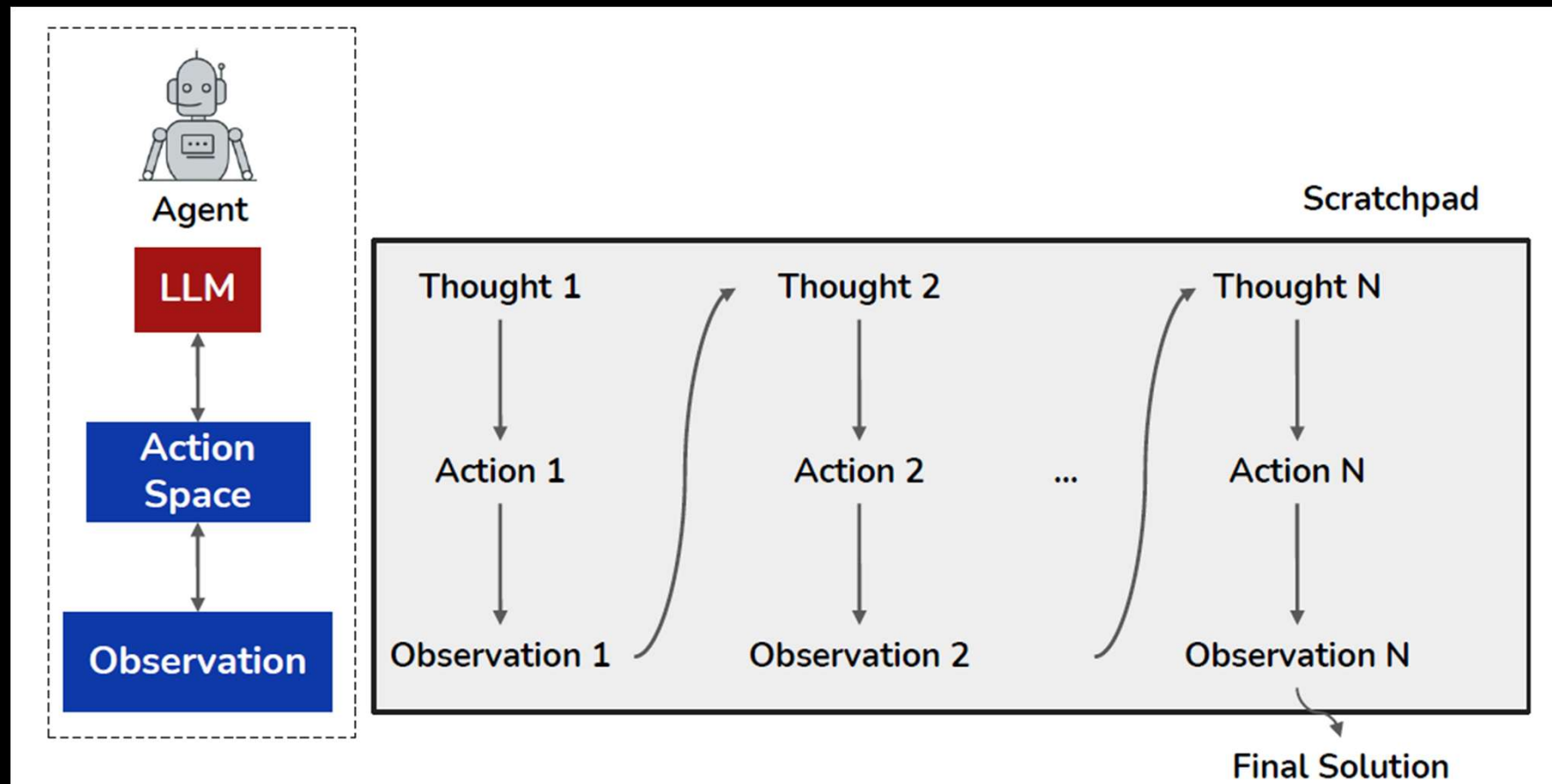
The average balance for customers who do not default is approximately 1552.84, while the average balance for customers who default is approximately -61.80.

Customers who default have a significantly lower average balance compared to customers who do not default.

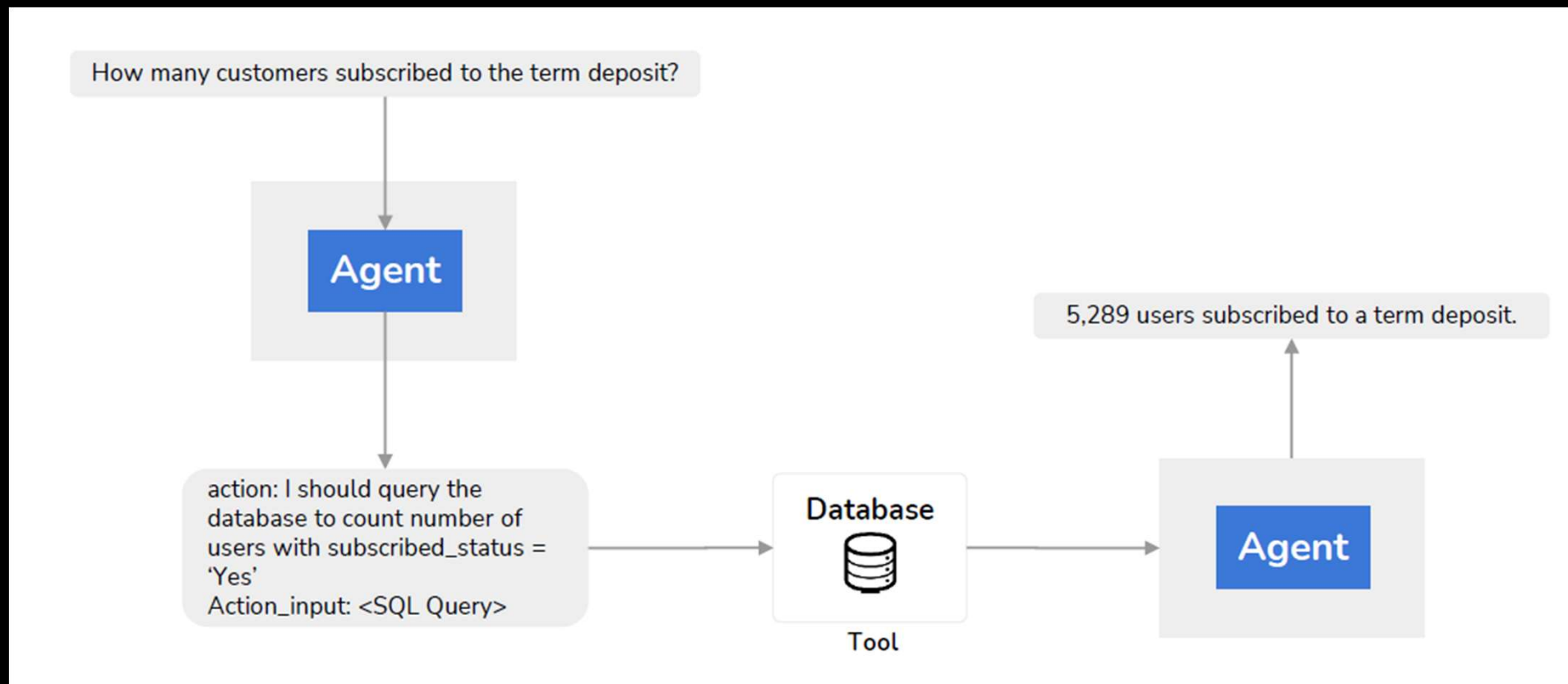
# Function calling and Agents



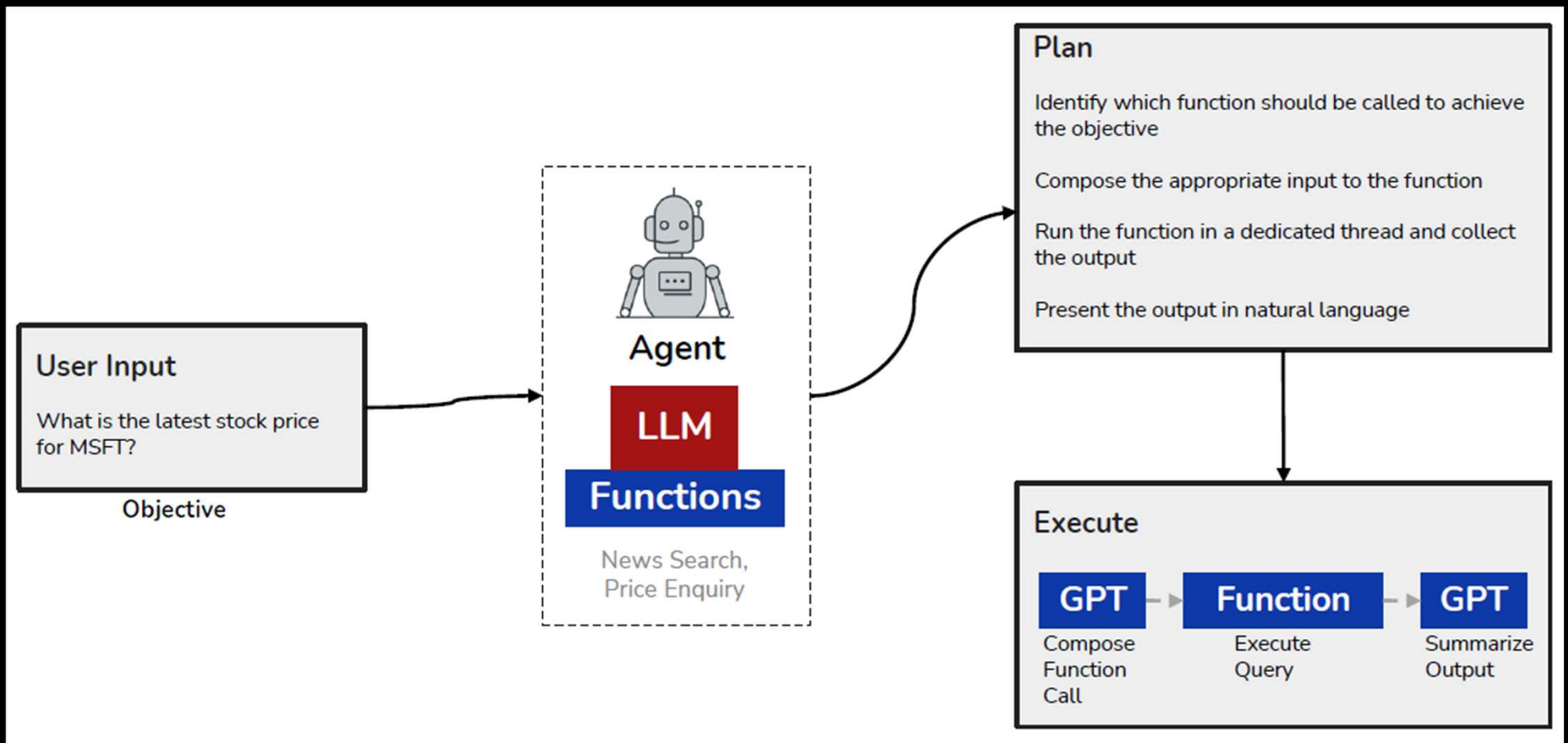
# ReAct paradigm (Reasoning and Action)



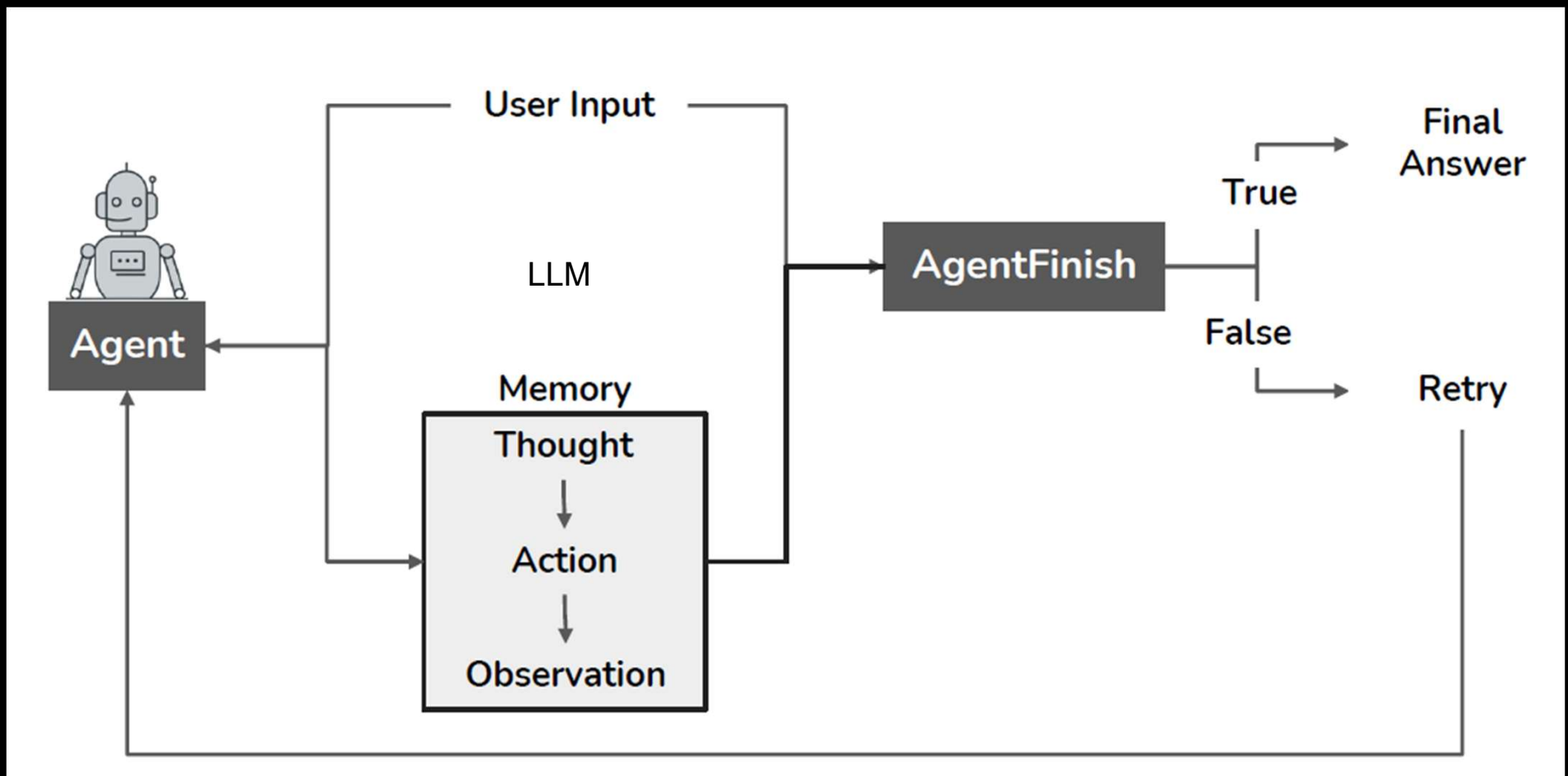
# Agent workflow



# Implement an Agents with function calling

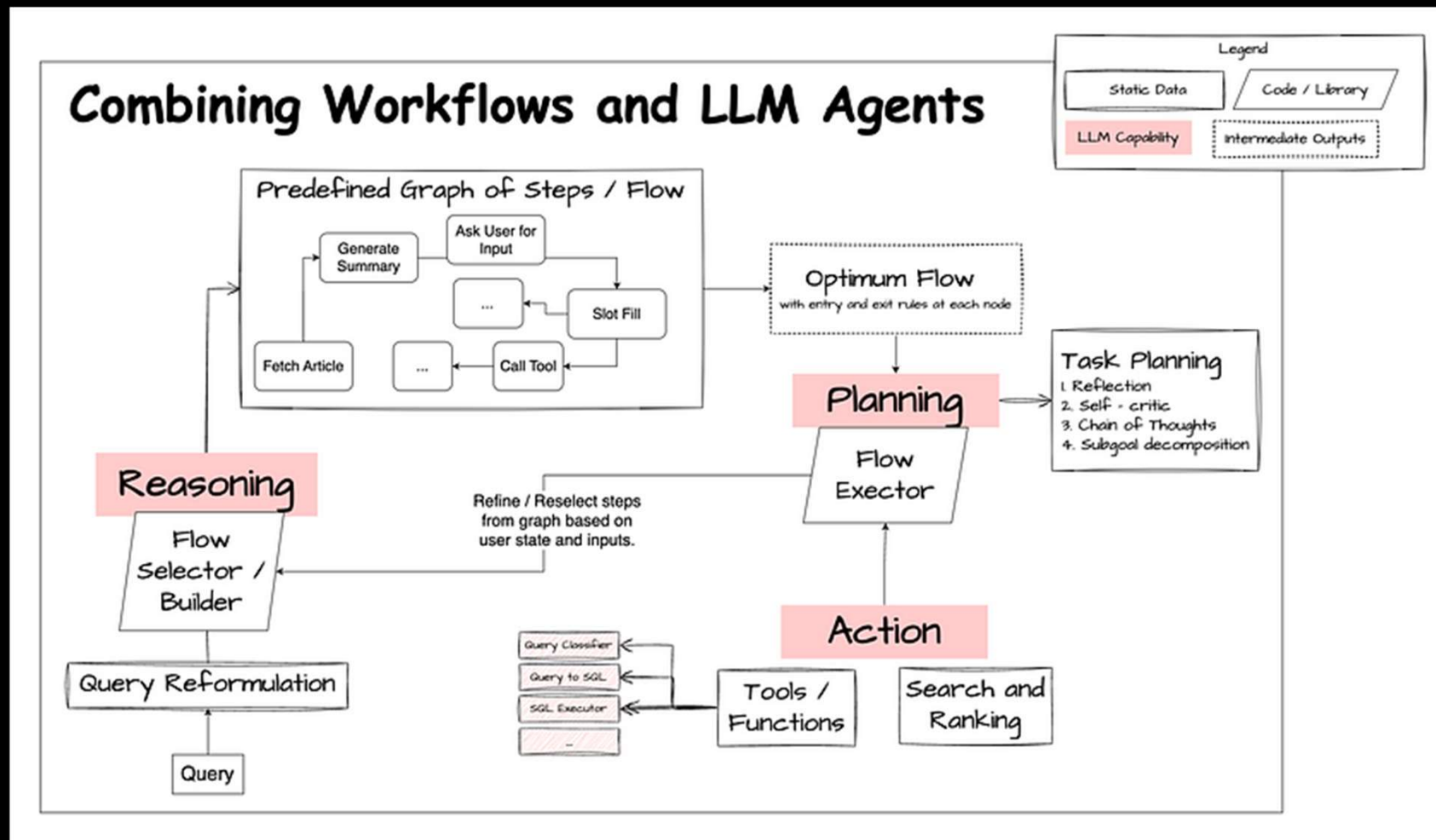


# Implement an Agentic workflow





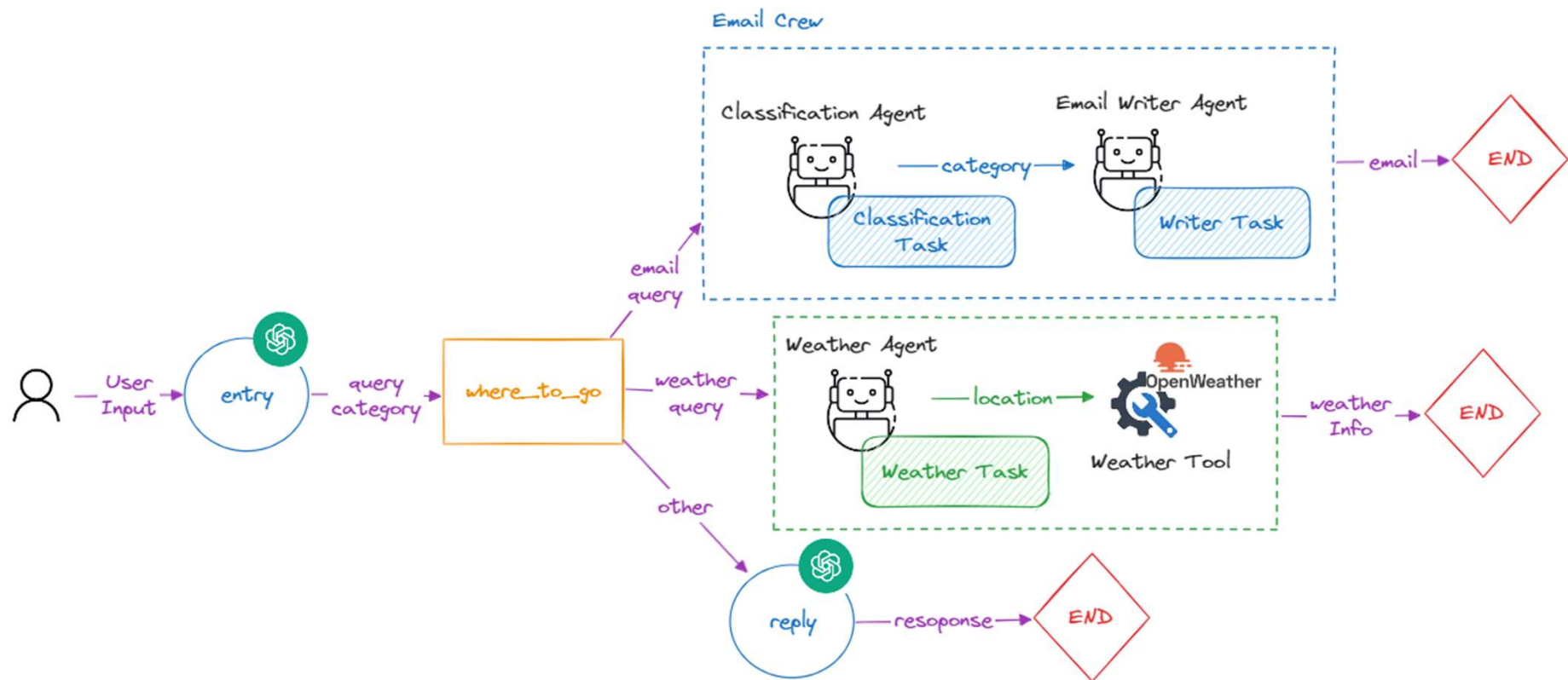
# Workflow with LLM Agents



<https://medium.com/@raunak-jain/path-to-production-for-llm-agents-f0a9cafd2398>



# Workflow example



Function calling and Agents  
Demos and hands-on

# Frameworks to build Agentic Apps

- Langchain Agents and Langgraph
- LLamaIndex Tools
- Open AI Swarm
- Crew AI

And many more (RASA, AutoGen, SmolAgents etc.)

# Considerations

- Langgraph, Crew, Swarm offer powerful orchestration and workflow components
- Application design considerations will involve properly marrying these capabilities with LLM capabilities around summarization, generation, reasoning and planning
- Rule based workflows are perhaps better captured by hard wiring the flow using these tools for predictability and reliability

# LangGraph

## **Primary Functionality**

LangGraph is designed for workflow management with a focus on visual graph-based design of tasks and dependencies.

## **Customization**

Offers highly customizable workflows through LangChain's API integrations, making it suitable for managing multiple tools and scenarios simultaneously.

## **Workflow Management**

Provides a visual graph-based approach, which is beneficial for complex task management and structured outputs.

## **Ease of Use**

Requires knowledge of LLM workflows and visual programming, which can be challenging for beginners.

# Crew AI

## **Primary Functionality**

CrewAI is built around the concept of a "crew" of agents working together. It uses a structured process flow to execute tasks sequentially or in parallel

## **Customization**

Offers role-based specialization, where agents are defined by roles and goals. This framework abstracts complexity by providing high-level constructs for task management.

## **Workflow Management**

Uses a sequential execution model by default, but can be scaled by running multiple crews in parallel.

## **Ease of Use**

Known for excellent documentation and a robust community, making it easier for users to get started

# OpenAI Swarm

## **Primary Functionality**

Swarm is a minimalist framework for multi-agent orchestration. It extends the OpenAI Chat Completions API and focuses on dynamic task distribution among AI agents

## **Customization**

Offers no-code customization through GPTs for specialized AI models. Agents are defined by system prompts and available functions.

## **Workflow Management**

Enables dynamic task distribution without a central controller. Control flows implicitly via agent handoffs and function calls

## **Ease of Use**

More beginner-friendly due to its simplicity and no-code customization options, but it requires manual management of logic



Demo of sample applications with LangGraph, Swarm,  
Crew AI